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# Building a searchable online corpus of Australian and New Zealand aligned speech

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**ABSTRACT**
Advances in automatic speech recognition technology, increases in bandwidth availability, and the widespread use of video streaming and sharing platforms have opened new horizons for corpus phonetics. CoANZSE Audio, a searchable online version of the Corpus of Australian and New Zealand Spoken English, provides access to over 195 million words of transcribed speech from transcripts of videos uploaded to YouTube by councils and other local government entities in Australia and New Zealand. Audio and forced alignment files are also available, making the resource suitable for the investigation of a range of research questions pertaining to morphosyntax, phonetics, and discourse. The resource, which is freely available via login through CLARIN, Europe's main language resources infrastructure network, was created through the use of open-source tools and software: yt-dlp, a Python library for collecting data from video and streaming websites; the Montreal Forced Aligner, a recent neural network alignment suite; and Parselmouth-Praat, Python bindings for the Praat acoustic analysis software. The website is powered by BlackLab, which combines a powerful search engine based on Apache Lucene with an intuitive web frontend. CoANZSE Audio may be useful for the investigation of regional differentiation of language features, and with additional annotation, differences in feature use according to social or demographic groups. Recent applications have included studies of double modals, a rare syntactic feature, and apology sequences. The nature of the audio and alignment data may make the resource especially suitable for the study of regional phonetic variation. Furthermore, the methods used to create the resource may be of interest to researchers seeking to adopt a pipeline approach for the creation of specialized corpora from publicly available online content.

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**KEYWORDS**
Corpus linguistics; Automatic Speech Recognition; YouTube; corpus phonetics; online resources; Australian English

## 1. Introduction

In recent years, developments such as improvements in Automatic Speech Recognition (ASR) technology, standardization of internet protocols for content streaming, and

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This article was originally published with errors, which have now been corrected in the online version. Please see Correction (http://dx.doi.org/10.1080/07268602.2024.2408086) increases in bandwidth have led to the widespread availability online of video, audio, and ASR transcript data, opening new perspectives for large-scale and corpus-based studies of naturalistic speech. Despite the proliferation of speech data online, however, relatively few structured language resources have been developed in which transcripts, audio data, and alignments have been made freely accessible to researchers.

This paper presents CoANZSE Audio (https://coanzse.org), a searchable web corpus of 195 million words from transcripts of videos uploaded to the YouTube channels of local government entities in Australia and New Zealand: local councils.[1] The resource, an updated version of the Corpus of Australian and New Zealand Spoken English (Coats, 2022a), includes audio and forced alignments for almost all of the transcript content, much of which is naturalistic and "ecologically valid" speech. The resource may be suitable not only for the investigation of lexical, morphosyntactic, and discourse properties of interaction, but also, as it provides access to a large repository of structured naturalistic audio data and alignments, phonetic analyses of contemporary Australian and New Zealand Englishes, especially in regards to geographical, discourse, and social aspects of phonetic variation. In total, the website provides access to over 18 million audio and alignment files and over 600 million individual force aligned phones, making it the largest currently available searchable online corpus and audio resource for these varieties of English.

From the perspective of corpus creation and language resource infrastructure, CoANZSE Audio represents an example of data harvesting and re-use for linguistic research purposes according to FAIR principles (findability, accessibility, interoperability, and reusability; Wilkinson et al., 2016): it is freely accessible to students and academics via authentication through CLARIN,[2] Europe's largest language resource infrastructure. In this respect, the methods used to create the corpus and the website may be relevant for similar large-scale data re-use and resource infrastructure projects in a linguistic context.

The article is organized as follows: in the next section, a brief overview is provided of ASR and forced alignment tools, followed by a discussion of existing comparable resources for the study of Australian and New Zealand Englishes. Section 3 provides details about the creation of the corpus and the CoANZSE Audio website, including information about the methods used to identify suitable sources of data, the scripts and procedures used for the collection of transcript and audio files, and the forced alignment of the data. Infrastructure issues such as data storage and backup, search engine implementation, data accessibility and authentication, and the services and technologies used to create the website are also discussed, as is, briefly, the legal framework for re-use of similar data for research purposes. In §4 potential applications for the corpus and its caveats and limitations, of which there are several, are discussed. The paper concludes with an outlook on future developments for the corpus and for the coanzse.org website.

## 2. Related work

### 2.1 ASR and forced alignment

Large-scale corpus phonetic analyses can require time-consuming and expensive manual transcription of audio recordings, but in recent years, the accuracy of ASR and forced

[1]This article is a modified and expanded version of Coats (2024a).
[2]https://clarin.eu.

alignment algorithms has greatly increased. Tools and services such as FAVE-Extract (Forced Alignment and Vowel Extraction; Rosenfelder et al., 2022), WebMAUS (Munich Automatic Segmentation; Kisler et al., 2017), or DARLA (Dartmouth Linguistic Annotation; Reddy & Stanford, 2015) can automate phonetic analysis workflows, and pipelines for automatic retrieval, alignment, and phonetic analysis of online content are increasingly used (e.g. Coats, 2023c).

Although ASR and forced alignment are prone to errors, automated phonetic processing pipelines based on ASR and forced alignment tools can reliably detect large-scale phonetic phenomena such as chain shifts in vowel quality that are limited to a particular geographical area. Coto-Solano et al. (2021), for example, compared ASR transcripts with manual transcripts of the same recordings in terms of how well vowels extracted on the basis of forced alignments showed the Southern Vowel Shift and the Northern Cities Shift, two chain shifts in the vowels of North American English. They found that, for the most part, vowel spaces generated by ASR transcripts matched the spaces generated from the manual transcripts in terms of the locations of particular vowels in F1/F2 formant space. Despite ASR errors, in large-scale studies in which many vowel tokens are extracted, "errors in the transcription may not affect the overall goal of producing vowel formants that are generally representative of a speaker's dialect features" (p. 3).

Forced alignment tools do not necessarily need to be trained on audio that matches the data to be analyzed. MacKenzie and Turton (2020) used FAVE-Extract and DARLA to align six samples of regional British English, finding them to work well even with American English acoustic models; they noted that "we have full confidence in recommending that researchers who work on non-American and non-standard varieties of English use these tools for forced alignment" (2020, p. 11). Meer (2020) compared FAVE-Extract, MAUS, and the Montreal Forced Aligner (McAuliffe et al., 2017) for alignment of English from Trinidad, finding that although they generally worked well, some aligners had problems with vowel realizations characteristic for Trinidad English. For Australian English, Gonzalez et al. (2020) found the "out-of-the-box" American English acoustic model provided by the Montreal Forced Aligner can be used to create accurate alignments for Australian English.

Overall, developments in the accuracy of ASR and forced alignment algorithms have aroused considerable interest among linguists and phoneticians, and the incorporation of these tools into processing pipelines has opened up new kinds of data to researchers, enabling the creation of big-data resources such as CoANZSE Audio.

### *2.2 Comparison with existing resources*

In recent years, coordinated efforts between researchers and institutions have resulted in the creation of infrastructure projects that facilitate the sharing of and access to data, including in Australia. These infrastructures include the Alveo Virtual Lab (Cassidy et al., 2014),[3] established as an online resource to host language corpora, the Language Data Commons of Australia, LDaCA,[4] an initiative of several Australian universities within the context of the Australian Research Data Commons (ARDC),[5] as well as initiatives such

[3]https://www.alveo.edu.au/.
[4]https://www.ldaca.edu.au/.
[5]https://ardc.edu.au/.

as the Language Technology and Data Analysis Laboratory (LADAL),[6] and the Australian Text Analytics Platform (ATAP).[7]

The linguistic properties of Australian and New Zealand varieties of English have been extensively investigated on the basis of empirical data, and several resources containing transcribed speech and/or audio are available. The Australian component of the International Corpus of English (Greenbaum & Nelson, 1996; Smith & Peters, 2023) contains approximately 600,000 words of transcribed speech collected in the years 1992–1995 from various genres. Some of the transcript data, but not audio recordings, are available through LDaCA. Other Australian transcript corpora include the Monash Corpus of Spoken Australian English (Bradshaw et al., 2010) and the Griffith Corpus of Spoken Australian English (Haugh & Chang, 2013), both of which focus on particular locations (Melbourne for the Monash Corpus and Brisbane for the Griffith Corpus) and which are included in the Australian National Corpus. Transcripts from these corpora are planned to be incorporated into the LDaCA infrastructure, but as of late 2023, it is not clear if audio files and/or alignments will be included. An additional resource of historical and linguistic importance is the Mitchell and Delbridge recording archive (Mitchell & Delbridge, 1998), consisting of spontaneous narratives, sentences, and word lists read by secondary school students at 327 schools in all Australian states (with the exception of the Northern Territory). A significant number of the recordings were digitized in 1998 and are now available through the LDaCA infrastructure. Location information and other metadata are available in the "speaker" metadata field but are not directly searchable through the LDaCA search interface. The AusTalk resource (Cassidy et al., 2017; Estival et al., 2014) is a database of recordings of spoken Australian English collected from 2011 to 2013 from more than 900 persons in 15 different locations in Australia. The resource includes recordings of read word lists as well as conversational speech, some of which have been transcribed. As of late 2023, plans have been made to provide access to AusTalk through LDaCA. Several current research initiatives are focused on the collection of audio data for specific varieties of Australian and New Zealand Englishes. Noteworthy is the Sydney Speaks project, which consists of data from several projects from the 1970s/1980s and 2010s/2020s (Travis et al., 2023; Travis, 2024) and comprises over 130 hours of recordings and 1.25 million words of transcript data. Although several corpora of Australian English are available, relatively few resources document naturalistic speech outside of major Australian urban areas.

For New Zealand, important resources of transcribed speech include the spoken component of the New Zealand International Corpus of English (ICE-NZ), with approximately 600,000 word tokens from the 1990s; the Wellington Corpus of Spoken New Zealand English (Holmes et al., 1998), comprising 1 million words and also recorded mainly in the 1990s; and the Origins of New Zealand English project (ONZE; Gordon et al., 2007), which includes recordings from the 1940s created by New Zealand's National Broadcasting Service as well as interviews conducted by academic researchers in the late twentieth and early twenty-first centuries. ONZE material includes more than 1,000 hours of speech from more than 600 speakers and 2 million word tokens. The UC QuakeBox Corpus (Clark et al., 2016; Walsh et al., 2013) comprises 576 recordings of New Zealanders recounting

[6]https://ladal.edu.au/index.html.
[7]https://www.atap.edu.au/.

their experiences of the 2010 and 2011 Christchurch earthquakes. The recordings are available via a website,[8] and researchers can request access to transcripts and alignments. ONZE data are not yet widely available to the research community in an online searchable format.

Resources similar to CoANZSE, using the same sampling procedures, have been created for English in North America (CoNASE) and Great Britain and Ireland (CoBISE; both Coats, 2023a). The Australian and New Zealand corpus thus represents a potential data source for comparisons of English speech with other major national varieties such as American, Canadian, British, and Irish English. The next section describes the creation and structure of CoANZSE Audio, a resource that may provide data suitable for this kind of analysis.

## 3. Corpus description

### *3.1 CoANZSE overview*

The creation of the resource required multiple data collection, processing, and preparation steps. A schematic overview of the corpus creation process, with the month and year in which the steps were undertaken, is shown in Figure 1.

Many councils in Australia and New Zealand maintain a YouTube channel to which they upload recordings of meetings, promotional videos, news, or other types of content. The first step in corpus creation was therefore the identification of YouTube channels of local government bodies in Australia and New Zealand. Starting from lists of councils in Australian states/territories and in New Zealand, all links to YouTube channels were collected by a custom script building on the open-source Python library yt-dlp;[9] these links were then manually verified and all available transcripts were downloaded from 482 channels (see Coats, 2023a and Coats, 2022a for details). Transcript data were downloaded during the first half of 2022; conversion to part-of-speech tagged and geolocated transcripts was done with custom scripts utilizing the Python libraries SpaCy (Montani et al., 2021) and geopy (Esmukov et al., 2018).[10] In August 2022 CoANZSE was made available for download in tabular format.[11]

The speech in the resource represents a range of different scripted and organic/naturalistic registers, including official and unofficial meetings of local government bodies, news and informational broadcasts about local communities, announcements, interviews with political, cultural, and other local figures, recordings of cultural events such as concerts or festivals, videos made for children, and a number of other content types.

The data in the corpus are recent: Figure 2 shows the number of transcripts by year of upload to YouTube. Beginning in 2020, some of the recordings are of virtual meetings conducted via conferencing software due to the Covid-19 pandemic.

Sampling locations for Australia and New Zealand are shown in Figures 3 and 4.

Sampling was undertaken for channels in Australian states, the Northern Territory, and the Australian Capital Territory, as well as the Indian Ocean territories of Christmas Island

[8]https://quakestudies.canterbury.ac.nz/.
[9]https://github.com/yt-dlp.
[10]https://spacy.io, https://github.com/geopy.
[11]https://doi.org/10.7910/DVN/GW35AK.

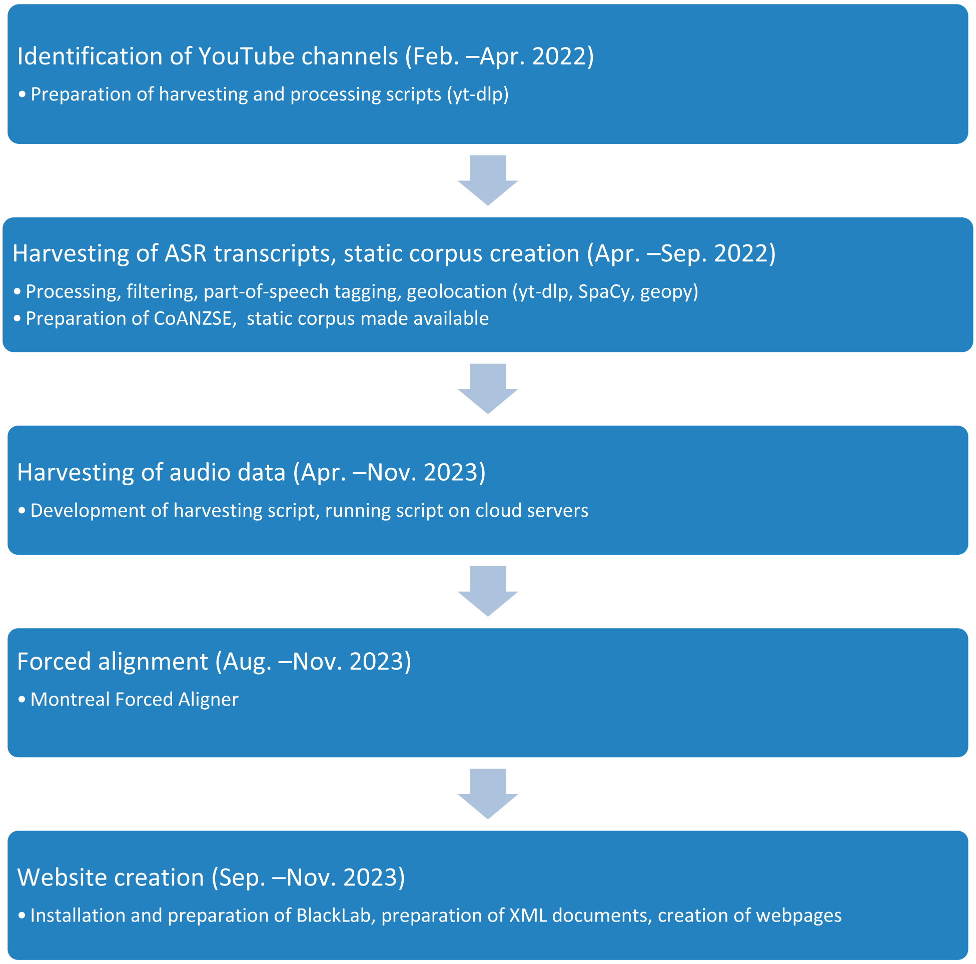


**Figure 1** CoANZSE Audio creation steps

and the Cocos (Keeling) Islands. For New Zealand, data were collected from 74 channels of councils in 65 locations throughout the country. An overview of the state or territory-level subcorpora in terms of channels, transcripts, word tokens, video length, and audio/TextGrid files is provided in Table 1.

The speech recorded in the transcripts and audio files is from almost 56,000 videos, but the videos (and hence transcripts) do not contain information about individual speakers (see §4.2). The exact number of different speakers in the corpus is unknown but is likely to be between 50,000 and 100,000.[12]

The data in the videos, audio files, transcripts, and forced alignments were collected from the public outreach channels of government bodies in Australia and New Zealand, mainly local councils, and represent public speech such as recorded council

[12]An estimate; the number could be estimated more accurately by applying speaker recognition models to the underlying audio recordings. See, e.g. https://huggingface.co/pyannote/speaker-diarization.

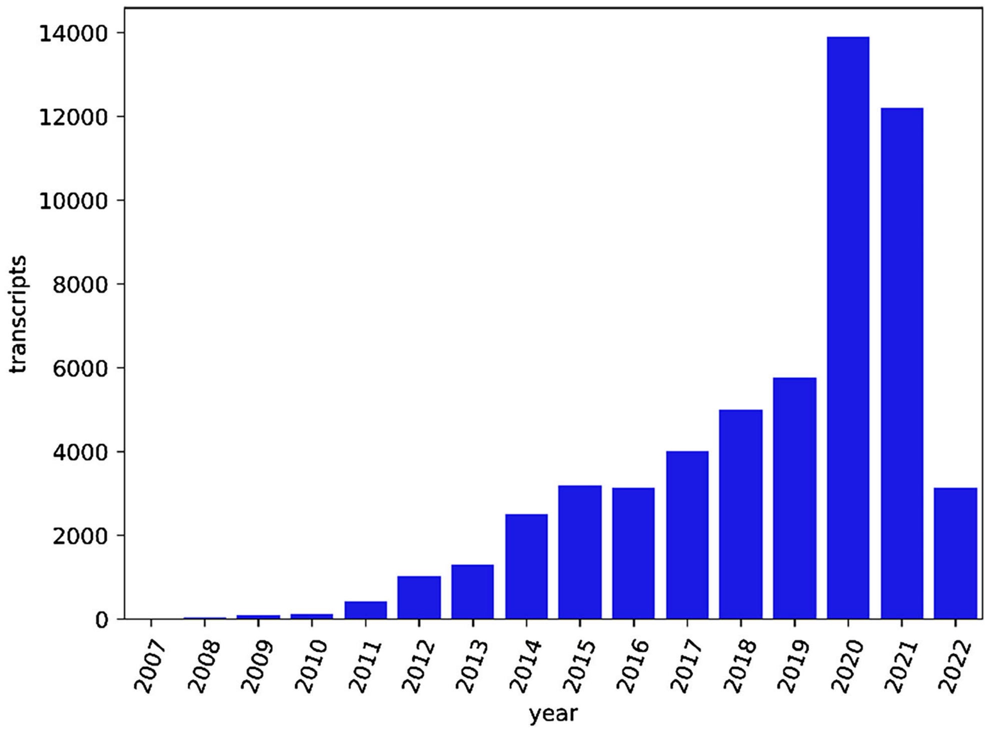


**Figure 2** Number of transcripts by upload year

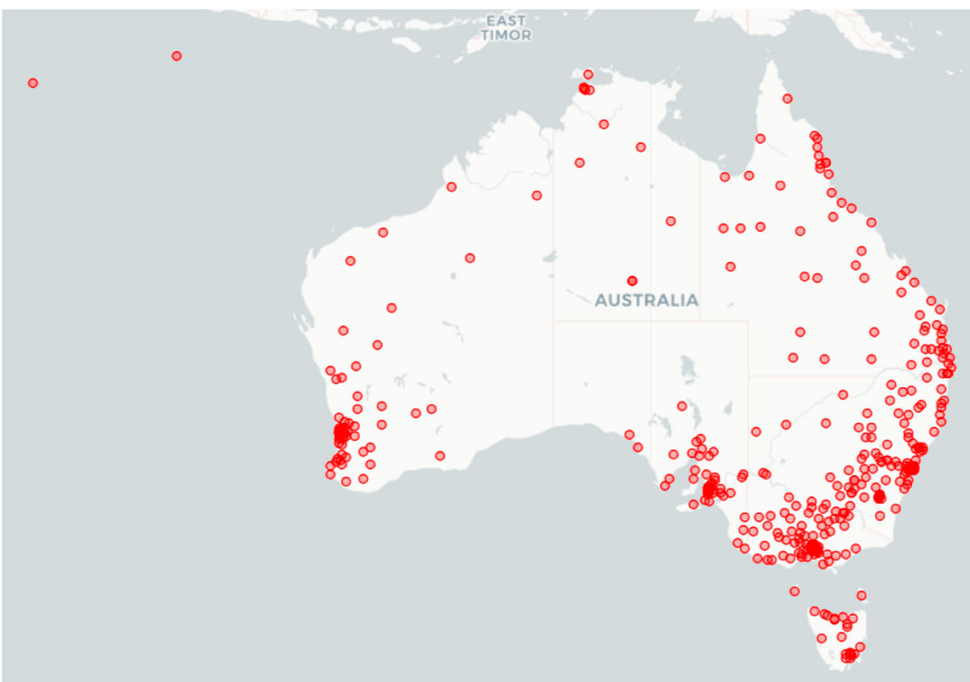


**Figure 3** Australian sample locations. (Map images © OpenStreetMap contributors, © CartoDB.)


**Figure 4** New Zealand sample locations. (Map images © OpenStreetMap contributors, © CartoDB.)

meetings, intended to be broadcast for the benefit of citizens, rather than private communication.

Sensitive personal information of the kind that requires special consideration according to Australian and New Zealand laws, such as medical or health records, political party membership, or sexual behaviour, is unlikely to be included in the transcript or audio data retrieved from YouTube, given the nature of the business of local councils. As councils in Australia and New Zealand are also obliged to restrict access to sensitive personal data, it is unlikely that they have produced and uploaded to YouTube videos whose transcripts and audio include this kind of information. Nevertheless, coanzse.org explicitly recognizes

**Table 1** Size of subcorpora by administrative territory

| Country | State/ territory | Number of channels sampled | Number of transcripts | Subcorpus length in words | Length of corresponding videos in hours | Number of audio files | Number of TextGrid files |
|---|---|---|---|---|---|---|---|
| AUS | Australian Capital Territory | 8 | 650 | 915,542 | 111.79 | 41,752 | 41,253 |
| | New South Wales | 114 | 9,741 | 27,580,773 | 3,428.87 | 1,299,949 | 1,272,673 |
| | Northern Territory | 11 | 289 | 315,300 | 48.72 | 6,628 | 6,471 |
| | Queensland | 58 | 7,356 | 19,988,051 | 2,642.75 | 950,084 | 837,297 |
| | South Australia | 50 | 3,537 | 13,856,275 | 1,716.72 | 643,866 | 629,524 |
| | Tasmania | 21 | 1,260 | 5,086,867 | 636.99 | 240,453 | 233,910 |
| | Victoria | 78 | 12,138 | 35,304,943 | 4,205.40 | 1,624,830 | 1,595,737 |
| | Western Australia | 68 | 3,815 | 8,422,484 | 1,063.78 | 386,898 | 377,016 |
| NZ | | 74 | 18,029 | 84,058,661 | 10,175.80 | 3,926,216 | 3,809,449 |
| **Total** | | **482** | **56,815** | **195,528,896** | **24,030.82** | **9,122,676** | **8,803,330** |

individuals' rights to rectification, to erasure, and to be forgotten, and will anonymize or remove records at the request of individuals identified in the data.

### *3.2 Extraction of audio data*

Much more time-consuming than transcript collection was the downloading of audio data, which are substantially larger in size than text data. Scripts for audio download, primarily based on functions from yt-dlp as well as the open-source software ffmpeg,[13] were tested in early 2023 and implemented beginning in April 2023. Data collection scripts were run in parallel on servers hosted by Finland's Centre for Scientific Computing (CSC), typically on between three and seven virtual machines, with occasional intercessions necessary due to, for example, unexpected server shutdowns on the part of CSC for server maintenance, unexpected disruptions or changes to YouTube's service, the necessity to update software libraries (yt-dlp) to adapt to changes in streaming protocols, shutdowns due to storage capacity or local memory limitations, and shutdowns for unknown reasons. Australian audio was collected from April until August 2023; New Zealand audio from July until November 2023. Transcripts and audio were divided into 20-word chunks to mitigate the potential of cascading errors in the forced alignments as well as to allow short audio snippets, usually in the range of 5–10 s in length, to be served with search results on the website. For each transcript chunk, the corresponding audio file was cut based on the word timing tags on the first and last words of the chunk.

The data in the corpus are sampled under a range of recording conditions. Some councils use professional recording equipment for the production of videos and check microphone settings to ensure good speech fidelity. Others use more improvised setups and are less concerned with ensuring high production values for the resulting videos. The corpus thus contains a range of video and audio quality for the sampled recordings.

[13]https://ffmpeg.org.

YouTube servers (and the servers of other platforms that host video content) host video and audio files in multiple versions with different sizes. The technical protocols used for streaming, DASH (Dynamic Adaptive Streaming over HTTP) and HLS (HTTP Live Streaming), transmit data at a rate that depends on available bandwidth. For high-bandwidth connections, high-quality video and audio will automatically be selected, whereas for connections with lower bandwidth, for example on some mobile devices, smaller video and audio files, with lower pixel resolution and less audio depth, will be streamed. For the data collection for this corpus, the data collection script selected the best available audio for each video, using an MPEG-4 AAC codec; files were stored in the .m4a format. The audio quality of a file downloaded using this approach depends on the properties of the file initially uploaded to YouTube by the corresponding channel, but for the most part, this procedure resulted in files with an average audio sample rate of 44.1 kHz. For cloud storage and for forced alignment using the Montreal Forced Aligner, audio files were converted to FLAC format using ffmpeg.

The transcripts in CoANZSE Audio were generated by YouTube's ASR module, which likely comprises an end-to-end pipeline including components such as a recurrent neural network, a specialized acoustic model, and a very large language model. Although the details of the YouTube ASR architecture and training data have not been made public, the application of large recurrent neural networks to language data has in recent years resulted in tremendous increases in accuracy and overall useability (Baevski et al., 2020; Chiu et al., 2017; Sainath et al., 2015).

ASR accuracy is constrained by factors such as the quality of the recording equipment and physical configuration of the recording setup, the acoustic properties of the audio signal, individual speaker phonetic traits including accent and prosody, as well as the vocabulary used in the recorded speech. In general, lower-quality recordings (i.e. with lower sample rates, in Hertz, and bit depth of less than 16) will exhibit higher error rates, as will recordings of speakers with non-standard accents or speech that contains lexical items that are not in the underlying acoustic and language models of the ASR system. Background noise also affects the accuracy of ASR.

Google's speech-to-text cloud service includes models for Australian English; as such, YouTube, which is owned by Google, likely also implements an Australian English model for recordings such as the ones indexed in the CoANZSE data. The accuracy of the ASR transcripts that comprise the CoANZSE corpus has been estimated, based on comparison of ASR with manual transcripts for videos for which both transcript types are available: Coats (2024b) found a mean word error rate of 14% for the ASR transcripts, a value higher than those of human-annotated transcripts (typically 5% or less), but which nonetheless may be suitable for many types of analysis.

### 3.3 Forced alignment

Audio and text segments were aligned with the Montreal Forced Aligner (McAuliffe et al., 2017), version 2.0, utilizing the default English acoustic model (english_us_arpa v2.0.0a), which was trained on a subset of the librispeech dataset (Panayotov et al., 2015). The output of the aligner, files in Praat's TextGrid format, show phones using ARPAbet notation, based on pronunciations recorded in the CMU Pronunciation Dictionary.[14] Australian data were aligned in September and October 2023; New Zealand data in October and November 2023. Not all transcripts in CoANZSE include audio and forced alignment data. Several videos were either deleted from YouTube or set to private between the time of collection of the transcripts (mid-2022) and the audio (mid-2023). Additionally, for some transcript files, the forced alignment pipeline failed to produce a usable TextGrid file, likely due to poor audio quality and/or inaccurate transcripts.

### 3.4 Website creation with BlackLab

The online, searchable version at https://coanzse.org provides access to the text, audio, and TextGrids. The search engine is a customized version of BlackLab (de Does et al., 2017),[15] a platform based on Apache Lucene and developed at the Dutch Language Institute in Leiden, Netherlands. The server and associated files (audio and TextGrids) are hosted by Finland's Centre for Scientific Computing.

For each corpus transcript, the following metadata fields are available: video title, channel title, country, state/territory (for Australia), name of council/organization responsible for the channel, channel URL, YouTube identifier,[16] upload date, video length (in seconds), transcript length (in number of word tokens), street address of the council/organization responsible for the channel, and latitude/longitude coordinates of that address. These metadata fields enable precise filtering according to location, as well as content-based filtering that may provide a basis for interpretations of discourse. For example, Figure 5 shows the proportion of transcripts that include the term "climate change" for each Australian state/territory and for New Zealand. As can be seen, the term is more likely to occur in New Zealand transcripts, followed by those from Tasmania and Victoria, and is least common in Northern Territory and Western Australia transcripts, possibly reflecting a greater importance associated with climate issues amongst New Zealand local councils, compared to Northern Territory and Western Australia councils.

#### 3.4.1 Hosting of audio and TextGrid files

The website provides access to almost 18 million audio and TextGrid files and more than 3.1 TB of data, requiring a distributed storage setup over 63 Swift object containers hosted by Finland's Centre for Scientific Computing. Indexing individual audio and TextGrid files to transcript excerpts was done via a script, allowing the CoANZSE Audio frontend to serve the correct audio and TextGrid files for every search hit. Audio files can be played in the browser.

#### 3.4.2 Access to CoANZSE Audio and data availability

The context for re-use of the underlying data is based on the "fair dealing" provisions of the Australian Copyright Act of 1968 and the New Zealand Copyright Act of 1994 as well as the "fair use" provision of American copyright law. Unlike some corpus resources,

[14] http://www.speech.cs.cmu.edu/cgi-bin/cmudict.
[15] https://inl.github.io/BlackLab/.
[16] YouTube videos are assigned unique 11-character identifiers which are part of the video's URL.

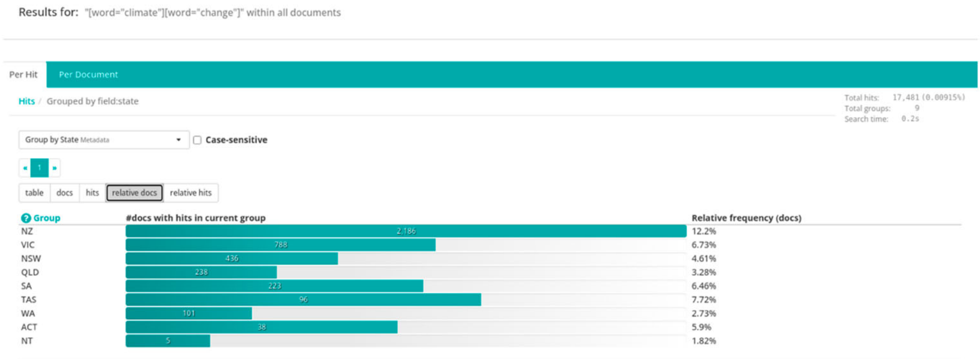


**Figure 5** Screenshot from coanzse.org

whose use may incur financial costs,[17] access to CoANZSE Audio is provided free of charge for academic and research purposes. The website can be accessed using authentication via the European CLARIN resource infrastructure, in accordance with the FAIR principles of open access and availability of research data.[18]

## 4. Corpus findings and implications

### *4.1 Applications of the corpus*

CoANZSE and the CoANZSE Audio website are new resources that can be used to investigate a wide variety of research questions in linguistic and computational social science contexts. A recent use case for CoANZSE data was to investigate the occurrence of a rare syntactic feature, the use of more than one modal verb in a single verbal phrase (e.g. *they would might consider it*). The feature, known as double (or multiple) modals, is a non-standard syntactic phenomenon traditionally thought to occur only in the speech of northern Britain and the southern US, as well as a few other locations (Montgomery & Nagle, 1994). Recent studies, however, have found that feature is more widespread in the UK, Ireland, and North America than previously thought (Coats, 2022b, 2023b), and can also be found in naturalistic speech in Australia and New Zealand (Morin & Coats, 2023). Thaler and Elsweiler (2023) used data from CoANZSE and the related CoBISE corpus to examine the association between gender and apology sequences in council meetings in New Zealand and the UK. They suggest that the formal situation of the meeting and the expectation of ritualized forms of politeness play a more important role than do gender or geographical location in these sequences.

The audio and alignment data from CoANZSE Audio may be suitable for the analysis of research questions in sociophonetics, including the extent to which phonetic realizations in Australian and New Zealand Englishes exhibit consistent regional divergence. Corpus data could be used to further investigate phonological and vocalic phenomena such as recent developments in the realization of short-front vowels in Australian English (Grama et al., 2019) or word-final *-er* (Grama et al., 2020).

[17] For example, the resources hosted by the Linguistic Data Consortium, the corpora available at english-corpora.org, or the resources available through the Sketch Engine website.
[18] https://force11.org/info/the-fair-data-principles.

As is well known, Mitchell and Delbridge (1965) found there to be relatively little regional variation in their 1959–1960 recordings, collected from almost all Australian schools, in terms of phonetic, syntactic, or discourse properties of Australian English: "Australia is, generally speaking, linguistically unified" (p. 13). The underlying data, however, digitized in 1998 and now available through LDaCA, have not yet been processed in a manner that would be suitable for a quantitative analysis of regional variation in phonetic (or other linguistic) features, although this is eminently possible. An analysis comparing regional pronunciation patterns of 1959/1960 with those of the 2020s, on the basis of CoANZSE data, may represent a topic for future investigation (cf. Cox et al., 2014).

### 4.2 Caveats and limitations

While the CoANZSE Audio resource can be used to investigate a potentially large number of research questions, there are several caveats and limitations pertaining to the data contained in the corpus.

A first consideration concerns the speech genres recorded in the transcripts and audio files. The videos uploaded to local council YouTube channels contain a variety of speech contexts, including recordings of public meetings, interviews, news announcements, promotional videos, and other types of content, but they cannot be considered representative for speech in general in Australia and New Zealand – many kinds of speech are poorly represented in the data, for example informal and unscripted speech between close friends or family members. Nevertheless, the size of the corpus and the wide range of speech types represented in the data make it likely that it captures much of the morphosyntactic, pragmatic, and phonetic variation of the underlying, abstract frames of "Australian/New Zealand English".

As noted in §3, ASR transcripts contain errors; the average WER of CoANZSE transcripts has been estimated to be 14%. As discussed by Coats (2023a), the potential for transcript errors needs to be taken into account in any analysis, particularly of low-frequency phenomena. The effect of transcript errors can be mitigated by checking the underlying audio signal, which can be done immediately for search hits in the CoANZSE Audio interface. For high-frequency phenomena, the signal in the data of accurately transcribed segments will be more robust, allowing valid inferences to be drawn.

The accuracy of the TextGrid files generated by the Montreal Forced Aligner has not been assessed via comparison with manually annotated alignments. However, Gonzalez et al. (2020) and MacKenzie and Turton (2020) suggest that the aligner produces good results for non-American varieties of English, including Australian English. Finally, YouTube ASR transcripts are not diarized, and contain no speaker demographic information, making analysis on the basis of traditional sociolinguistic or sociophonetic variables such as age, gender, or educational level difficult. While some demographic information can be manually or automatically annotated via inspection of the underlying videos, the resource is not well suited to out-of-the-box traditional sociolinguistic approaches in which age, gender, and other demographic variables are modelled.

Despite these caveats, the advantages of the corpus, especially in terms of size, ecological validity, and detailed location information, are likely to make it useful as a data source for a wide range of studies, especially those in which relatively large data samples are necessary.

### 4.3 Future developments and conclusion

The procedures used to create the CoANZSE Audio resource can be applied for the creation of specialized language corpora in general. CoANZSE Audio content is drawn from government YouTube channels, but corpora sampled from other kinds of content, such as channels exemplifying specific social groups or speech genres, could also be created. In terms of regionally defined varieties of English, an obvious next step for the corpus creation pipeline described in this article would be to create similar resources for varieties of English such as those used in India, Nigeria, South Africa, or the Caribbean.

As far as the content of CoANZSE Audio is concerned, the business of local councils is ongoing, and the YouTube channels from which CoANZSE Audio data have been sampled are continually updated with new videos. A future planned development for CoANZSE Audio is to modify the data collection and processing scripts in order to automate the addition of new data to the resource by automatically retrieving new content at regular intervals. In addition to this, improvements to scripts, for example the utilization of specifically Australian and New Zealand acoustic models for forced alignment, may improve the quality of the TextGrid files and hence the reliability of studies that use CoANZSE data.

To summarize, CoANZSE Audio represents a new kind of corpus: a fully searchable, fast, annotated text corpus which additionally provides access to hundreds of millions of words of aligned acoustic data. It is hoped that the website will provide a rich and useful source of data for a wide range of studies in the phonetics, linguistics, and discourse of contemporary English in Australia and New Zealand.

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No potential conflict of interest was reported by the author(s).

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